

Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Machine Learning Internship Project | Unified Mentor

Abstract

This project develops a machine-learning-based buyer segmentation and investment profiling system for real estate market intelligence. The workflow covers data cleaning, feature engineering, categorical encoding, feature scaling, K-Means clustering, hierarchical clustering, validation, interpretation, and Streamlit deployment. The final K-Means model uses three clusters and produces Standard Property Buyers, Premium Large-Property Buyers, and High-Value Multi-Property Buyers.

1. Background and Context

Real estate buyers differ in purchasing motivation, property preferences, financing patterns, geography, and satisfaction. Treating all buyers as one group can lead to generic marketing and less effective targeting. Buyer segmentation provides a data-driven approach to discovering groups with similar behavior.

2. Problem Statement

The system aims to discover hidden patterns among real estate buyers and create interpretable segments using investment purpose, geographic behavior, financing patterns, customer demographics, and property activity/value.

3. Dataset and Data Preparation

The analysis produced a final client-level dataset of 2,000 buyers. The cleaned source contained 2,000 client records and 10,000 property records, with zero duplicate client records and zero duplicate property records. The final modeling dataset contained 13 features before encoding and 84 features after categorical encoding.

4. Feature Engineering

The modeling features included client type, gender, country, region, acquisition purpose, loan behavior, referral channel, satisfaction score, age, total properties, total property value, average property price, and average area. Categorical variables were encoded and numerical variables were scaled for clustering.

5. Methodology

Data cleaning → feature encoding → feature scaling → K-Means evaluation → hierarchical clustering validation → cluster interpretation → Streamlit deployment.

6. Model Selection and Validation

K-Means silhouette scores for 2–8 clusters were 0.1593, 0.1661, 0.1182, 0.1006, 0.0963, 0.0926, and 0.0901 respectively. Three clusters had the highest tested silhouette score (0.1661), so the three-cluster K-Means model was selected. Hierarchical clustering with three clusters achieved 0.1315. PC1 and PC2 explained 42.71% of variance.

7. Final Buyer Segments

Cluster	Buyer Segment	Buyers	Buyer %	Avg Age	Avg Properties	Avg Property Value	Avg Area
0	High-Value Multi-Property Buyers	50	2.5%	64.28	7.32	\$2.42M	1,101.05
1	Premium Large-Property Buyers	852	42.6%	55.59	3.58	\$1.46M	1,347.41
2	Standard Property Buyers	1,098	54.9%	56.12	3.54	\$1.05M	994.46

8. Segment Insights

High-Value Multi-Property Buyers: 50 buyers (2.5%), average age 64.28, average 7.32 properties, average property value \$2,416,602.86, average satisfaction 3.52.

Premium Large-Property Buyers: 852 buyers (42.6%), average property area 1,347.41 sqft, average property price \$410,255.97, and average property value \$1,458,615.10.

Standard Property Buyers: 1,098 buyers (54.9%), average property value \$1,053,898.68 and average area 994.46 sqft.

The most common acquisition purpose, client type, and loan behavior across the clusters were Home, Individual, and No respectively.

9. Business Recommendations

Target High-Value Multi-Property Buyers with relationship-focused and premium investment campaigns. Position larger-area inventory and premium recommendations for Premium Large-Property Buyers. Use scalable, value-focused offerings for Standard Property Buyers. Combine segmentation with geographic and acquisition-purpose filters for more targeted decisions.

10. Streamlit Dashboard

The deployed dashboard provides KPIs, buyer segmentation, investment/property analysis, geographic analysis, segment insights, and filters for country, region, acquisition purpose, and client type.

11. Limitations

The final silhouette score of 0.1661 indicates limited cluster separation, so the segments should be considered exploratory decision-support groups rather than perfectly separated classes. Results are also limited to the variables present in the supplied dataset.

12. Conclusion

The project successfully applies unsupervised machine learning to real-estate buyer intelligence. The three K-Means segments reveal meaningful differences in property holdings, property value, price, area, age, and satisfaction, and the Streamlit dashboard makes these findings accessible for interactive analysis.

13. Project Deliverables

Google Colab analysis notebook; buyer_segmentation_final.csv; app.py; requirements.txt; deployed Streamlit dashboard.